Hypersensitivity reactions:

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SGL 6

Objectives:

- The students must learn
- All detail about the definition, types and immune mechanisms of the hypersensitivity reactions.

Hypersensitivity reactions Definition and types

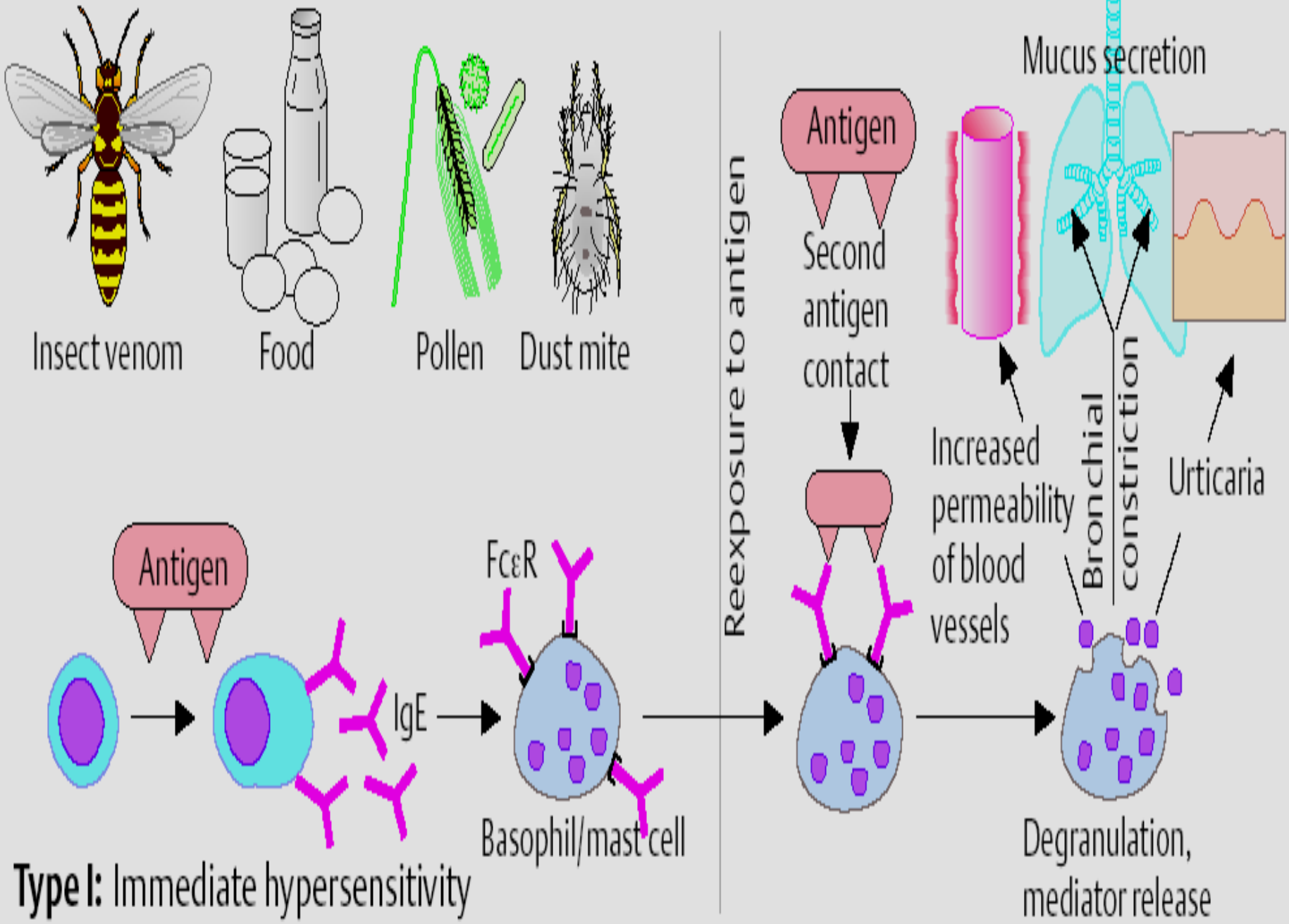
- It is a harmful immune response in which increased reactivity or sensitivity by the body to an antigen to which it has been **previously exposed**.
- Hypersensitivity responses can be produced through **endogenous** self-antigens or **exogenous antigens** as microbial and non-microbial (drugs, foods, pollens, chemicals, and dust)components.
- Hypersensitivity subdivided into **5 types:**
- Type I: Immediate or anaphylactic hypersensitivity
- Type II: Antibody-mediated cytotoxic hypersensitivity
- Type III: Immune complex-mediated hypersensitivity
- Type IV: Delayed type or cell-mediated hypersensitivity

Type I: Immediate or anaphylactic hypersensitivity

- It is an **allergic reaction** provoked by re-exposure to a specific type of antigen(allergen).
- The reaction may be either **local or systemic**.
- Symptoms vary from mild irritation to sudden death from **anaphylactic shock.**

Allergic reaction

- Exposure by ingestion, inhalation, injection, or direct contact.
- The allergen (pollen, animal dander, foods, and various drugs induces the formation of the IgE antibody which also called reagenic Ab.
- IgE has a high **affinity** for mast cells and basophils.
- **Reexposure** to the same allergen results in cross-link between the **antigen and IgE** on **mast cells** results in the **degranulation** of mast cells and the release of histamine and other pharmacologically active mediators such as **histamine, leukotriene, and prostaglandins** that act on the surrounding tissues. The principal effects of these products are **vasodilation** and **smooth-muscle contraction** within minutes.



Type I: Immediate hypersensitivity

The effect of mediators:

The mediators have two phase effects:

- In the **immediate phase**, the symptoms as **wheal, flare, and itching** appear rapidly due to certain mediators as histamine which is **preformed mediators**.
- The **late phase** occurs 6 hr. after exposure to the antigen due to certain mediators as leukotrienes are synthesized after the **influx of inflammatory cells** such as neutrophils and eosinophils. Such late-forming mediators cause symptoms as **erythema and indurations**.
- **Clinical manifestations** of type I hypersensitivity depend on
- Rout of the allergen entry.
- Location of the mast cells bearing the IgE specific for the allergen.

Mediators	Effects	Comments
Preformed mediators		
Histamine	Smooth muscle contraction (e.g., bronchospasm) Increased capillary permeability Increased mucus secretions Increased airway resistance Chemokinesis	Probably the most important mediator in anaphylaxis
ECF-A	Chemotaxis of eosinophils that control allergic reactions by releasing arylsulfatase B (inactivates SRS-A) and histaminase (inactivates histamine)	
Synthesized mediators		
PAF	Platelet aggregation and release of vasoactive amines Increased vascular permeability Smooth muscle contraction Bronchoconstriction Neutrophil chemotaxis	An acetyl glyceryl ether phosphoryl choline; inactivated by phospholipases
SRS-A leukotrienes LTC4, LTD4, LTE4	Smooth muscle contraction Increased capillary permeability Mucus secretion in airways	Lipoxygenase metabolite of arachidonic acid; probably a major factor in asthmatic bronchospasm
LTB4	Chemotaxis	
Prostaglandins PGE_1, PGE_2	Strong bronchodilation Strong vasodilation	Cyclooxygenase metabolites of arachidonic acid

ECF-A = eosinophil chemotactic factor of anaphylaxis; *LT* = leukotrienes; *PAF* = platelet-activating factor; *PG* = prostaglandin; *SRS-A* = slow-reacting substance of anaphylaxis.

Diagnosis:-

- **1-Family history**: hereditary predisposition to atopic allergies.
- **2-Skin test:** The purified form of allergen is injected intradermally and hypersensitivity reactions in the form of erythema and edema are observed **within 20-30 minutes**.
- **3-Basophile degranulation test** (measures release of histamine from leukocytes).

Diagnosis:-

- **4-Determination of serum level of IgE:**
- In **normal serum**, very **little IgE** presents.
- Normal levels in non-atopic individuals are usually **less than 20 IU/ml**.
- In atopic individuals generally **more than 20 IU/ml**, many have greater than 100 IU/ml.
- 1. **RIST** (radioimmunosorbent test) measures total IgE, the most sensitive of methods available.
- 2. **RAST** (radioallergosorbent test) quantities a specific IgE. Avoids exposing the patient to potentially dangerous skin testing.

Management:-

- **A-Environmental interventions:**
- 1-**Avoidance** of allergens.2-**Use of mask** and filter.
- **B-Pharmacological interventions:**
- 1-**Stabilization** of mast cell **membrane** by cromoglycan sodium.
- 2**-Prevention** of **inflammation** by corticosteroids.
- 3-**Symptomatic** as sneezing, itching and runny nose by antihistamines.
- 4-**Reverse** the **effects of mediators** by adrenaline and epinephrine.

- **C-Immunization interventions:-**
- 1-**Desensitisation or hyposensitization**:

Is a process by which a person is made **unresponsive** or **hyporesponsive** to a particular allergen.

- In this process, small doses of allergens are injected over a prolonged period of time with a gradual increase in dose. **IgG antibody** is produced which **blocks the Ag** by binding with it and does not allow the Ag to reach the mast cell-bound IgE.

- 2-**Using antibody to IgE**(monoclonal anti-IgE) to counteract and **remove** allergy causing **IgE**.

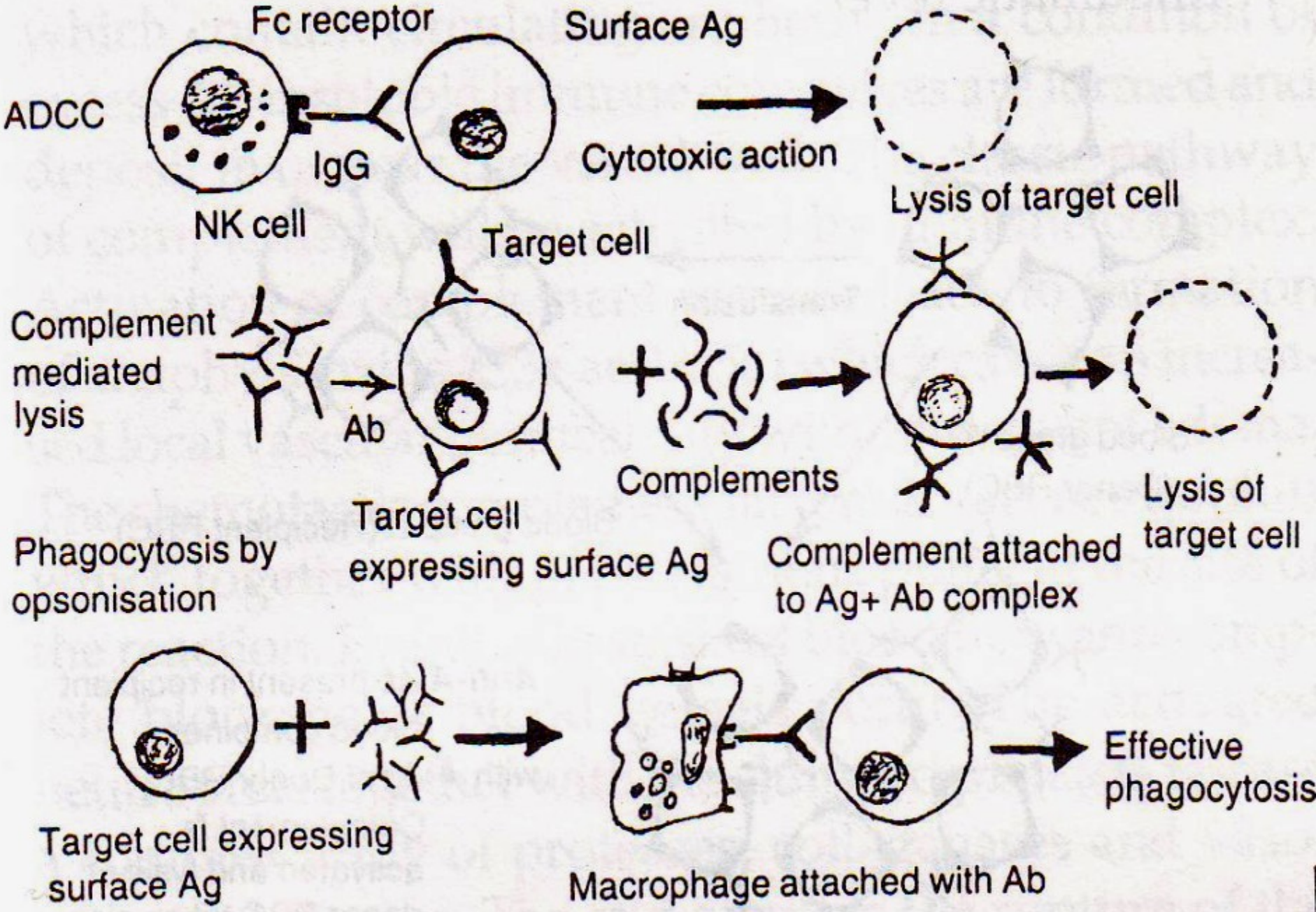
- 3-Use of modified allergens:-
- Modified allergens or **allergoids** are thought to stimulate the **suppresser T-cells** and thereby can **prevent anaphylactic** reactions.
- **Anaphylactoid reactions:-**
- Appear **clinically similar** to anaphylactic but **not IgE mediated**, the inciting agents usually **drugs** and directly induce the mast cells and basophils.

Examples of type I hypersensitivity:

- Allergic asthma
- Allergic conjunctivitis
- Allergic rhinitis ("hay fever")
- Anaphylaxis
- Angioedema
- Atopic dermatitis (eczema)
- Urticaria (hives)
- Eosinophilia

Type II: Antibody mediated cytotoxic hypersensitivity

- This responses are initiated by the toxic properties of antibody bind to antigens on the cell surfaces.
- The antigens may either be intrinsic (“self” antigen) or extrinsic.
- The reaction takes hours to a day.
- **Mechanisms are involved in cell damage in type-II reaction:**
- 1-Antibody-dependent cell-mediated cytotoxicity(ADCC). Antibodies are usually of IgG and / or IgM types.
- 2-Complement mediated cell lyses.
- 3-Immune adherence followed by phagocytosis.


Fc receptor
Surface Ag
ADCC
IgG
Cytotoxic action
Lysis of target cell
NK cell
Target cell
Complement mediated lysis
Ab
Complements
Lysis of target cell
Phagocytosis by opsonisation
Target cell expressing surface Ag
Complement attached to Ag+ Ab complex
Effective phagocytosis
Target cell expressing surface Ag
Macrophage attached with Ab

Conditions under which cytolytic reactions take place

- 1. Antigen **structural component** of a cell membrane or an **extrinsic antigen** that has attached to the cell.
- 2. Antigenic determinant on host cell that is identical or very similar to a foreign antigen.
- 3. New antigens as a result of drug-induced changes.
- 4. Antigen-antibody complexes that have formed in circulation and subsequently attached to red blood cells

Examples of type II hypersensitivity:

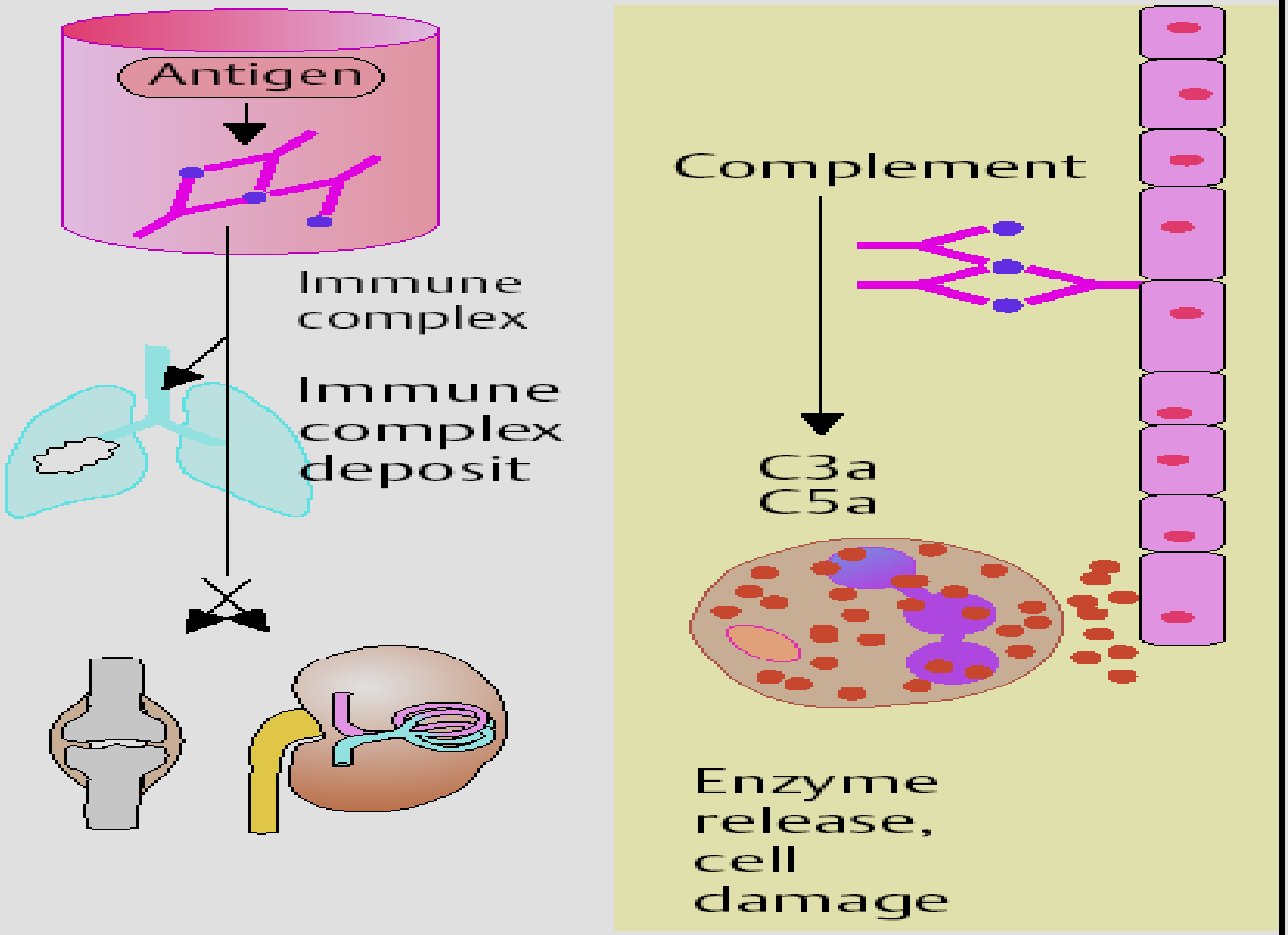
- 1-Mismatched **ABO blood** group transfusion and **Rh incompatibility** reactions.
- 2-Drug induced reactions: Chloramphenicol, phenacetin, and penicillin can **attach to surface proteins** on red blood cells and **initiate antibody formation** and result in **hemolysis.**
- 3- Infectious agents: Group A streptococci **cross-react** with **cardiac tissue** resulting in rheumatic fever and in Good pastures syndrome **antibodies** to basement **membranes of the kidneys and lungs**.

Examples of type II hypersensitivity:

- Autoimmune hemolytic anemia
- Goodpasture's syndrome
- Erythroblastosis Fetalis
- Pemphigus
- Pernicious anemia (if autoimmune)
- Immune thrombocytopenia
- Transfusion reactions
- Hashimoto's thyroiditis
- Graves' disease
- Myasthenia gravis
- Rheumatic fever
- Hemolytic disease of the newborn

Type III: Immune complex mediated hypersensitivity

- This type of reaction occurs when **insoluble Ag-Ab** complexes deposit in various tissues (typically the **skin, kidney, and joints**), activate the complement cascade, and **release** of certain products of complement as **C3a and C5a** which causes a local increase in **vascular permeability**.
- Chemotactic attraction of **neutrophils** to the site of the reaction cause induces an **inflammatory response** in the tissues.
- The reaction takes hours to days to develop.



Type III: Immune complex reaction

Types of Immune complex mediated hypersensitivity

1-Arthus reaction:

- Is the name given to the inflammation caused by the **deposition of immune complexes** at a localized site.

2-Serum sickness:

- Is a **systemic inflammation** response to the presence of i**mmune complexes deposited in many areas** of the body.

Arthus reaction

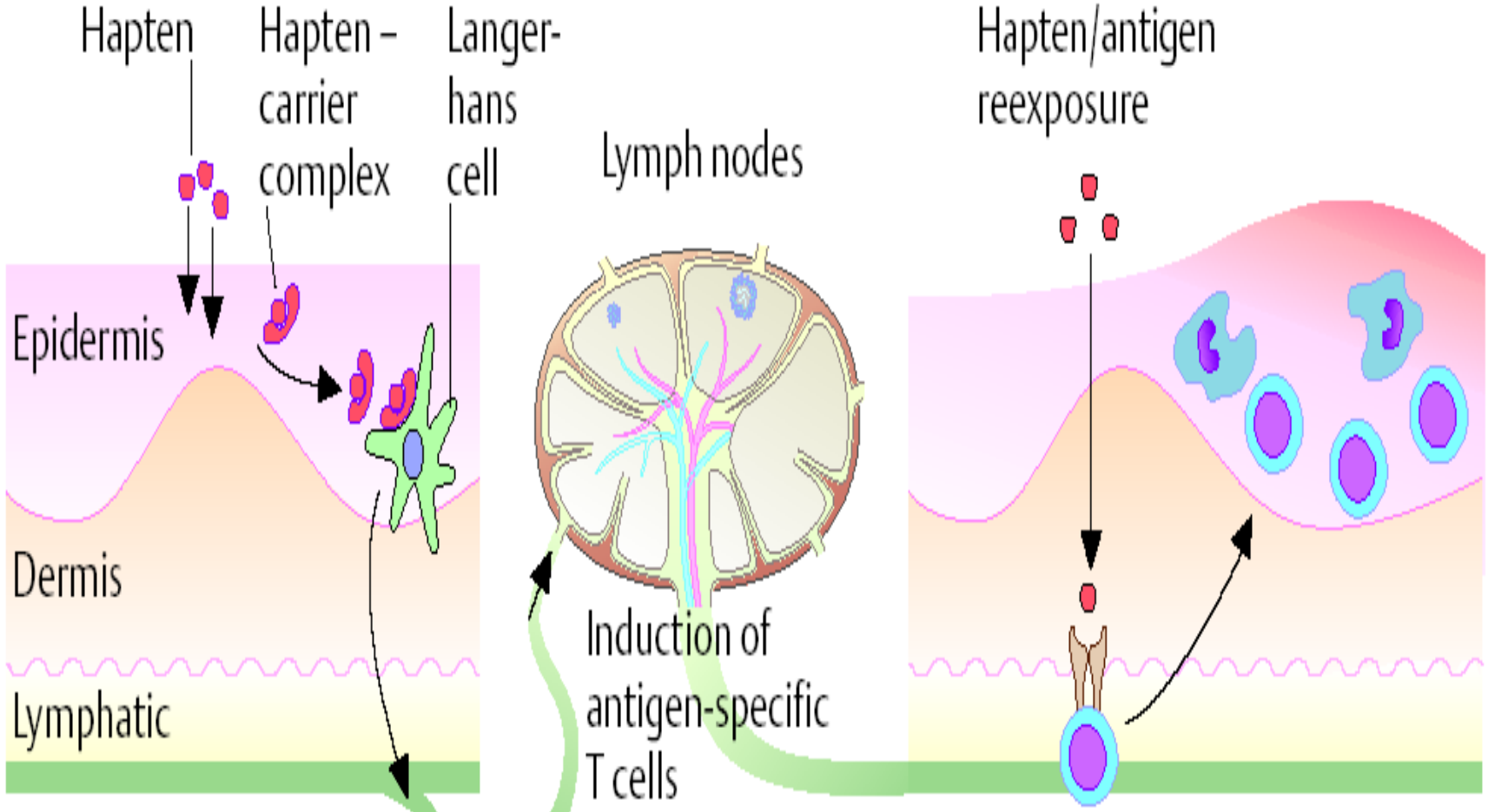
- The Arthus reaction was discovered by Arthus in 1903. Arthus repeatedly injected horse serum subcutaneously into rabbits. After four injections, he found that there was edema.
- An Arthus reaction is a **local vasculitis** associated with the **deposition of immune complexes** and activation of complement.
- Arthus reactions are characterized by **severe pain**, **swelling, induration, edema, hemorrhage, and occasionally by necrosis.**
- These symptoms and signs usually occur 4–12 hours after vaccination.

Examples of type III hypersensitivity:

- Immune complex glomerulonephritis
- Rheumatoid arthritis
- Serum sickness
- Subacute bacterial endocarditis
- Symptoms of malaria
- Systemic lupus erythematosus
- Arthus reaction
- Farmer's Lung (Arthus-type reaction)

Type IV hypersensitivity reactions :Cells mediated hypersensitivity

- It is called delayed-type as the reaction takes **two to three days** to develop.
- Unlike the other types, it is not antibody mediated but rather is a type of cell mediated response.
- **CD8+ cytotoxic T cells** and **CD4+ helper T cells** recognize antigen in a complex with either type 1 or type 2 major histocompatibility complex. The APC in this case is macrophages which secrete **IL-1,** that stimulates the proliferation of further CD4+ T cells.
- Transferred by sensitized T cells, not by serum.



Type IV: Delayed type hypersensitivity (DTH)

A. Types of hypersensitivity reactions

Type IV hypersensitivity reactions:

- Haptens as **poison oak** can penetrate the epidermis and bind to certain proteins in the skin (carrier proteins).
- Hapten-carrier complexes are bound by antigen presenting cells of the skin (**Langerhans cells**), which then migrate to regional lymph nodes. T-cell stimulation occurs at the lymph node. This is sensitization phase last 10-14 days.

Type IV hypersensitivity reactions:

- If the individuals are **re-exposed** to the hapten, **antigen–specific T cells** migrate to the skin, where they accumulate and proliferate.
- T helper also causes **edema formation and local inflammation** with the help of **cytokines**.
- The compound containing **nickel or chrome** and chemicals such as those found in rubber are typical triggers of type IV hypersensitivity reactions.

Examples of type IV hypersensitivity:

- Contact dermatitis (poison ivy rash)
- Temporal arteritis
- Symptoms of leprosy
- Symptoms of tuberculosis
- Transplant rejection
- Coeliac disease

Type V (Stimulatory Type) Hypersensitivity

- These reactions occur when IgG class antibodies directed towards cell surface antigens have a stimulating effect on their target.
- Antibodies combine with antigens on cell surface, which induces cells to proliferate and differentiate and enhances activity of effector cells.
- Type V HS reaction plays an important role in pathogenesis of Graves' disease, in which **thyroid hormones** are produced in excess quantity.
- It is postulated that **long acting thyroid stimulating antibody**, which is an **autoantibody** to thyroid membrane antigen, combines with thyroid-stimulating hormone (TSH) receptors on a thyroid cell surface.
- Interaction with TSH receptor produces an effect similar to the TSH, resulting in an excess production and secretion of thyroid hormone, which is responsible for Graves' disease.

References

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